Dr .Mohammed Harith

The AIM of this lecture is :

1 .Understand the anatomy of the gastrointestinal tract.

2 .Know the function & physiology of the gastrointestinal tract .

Intended learning outcomes :

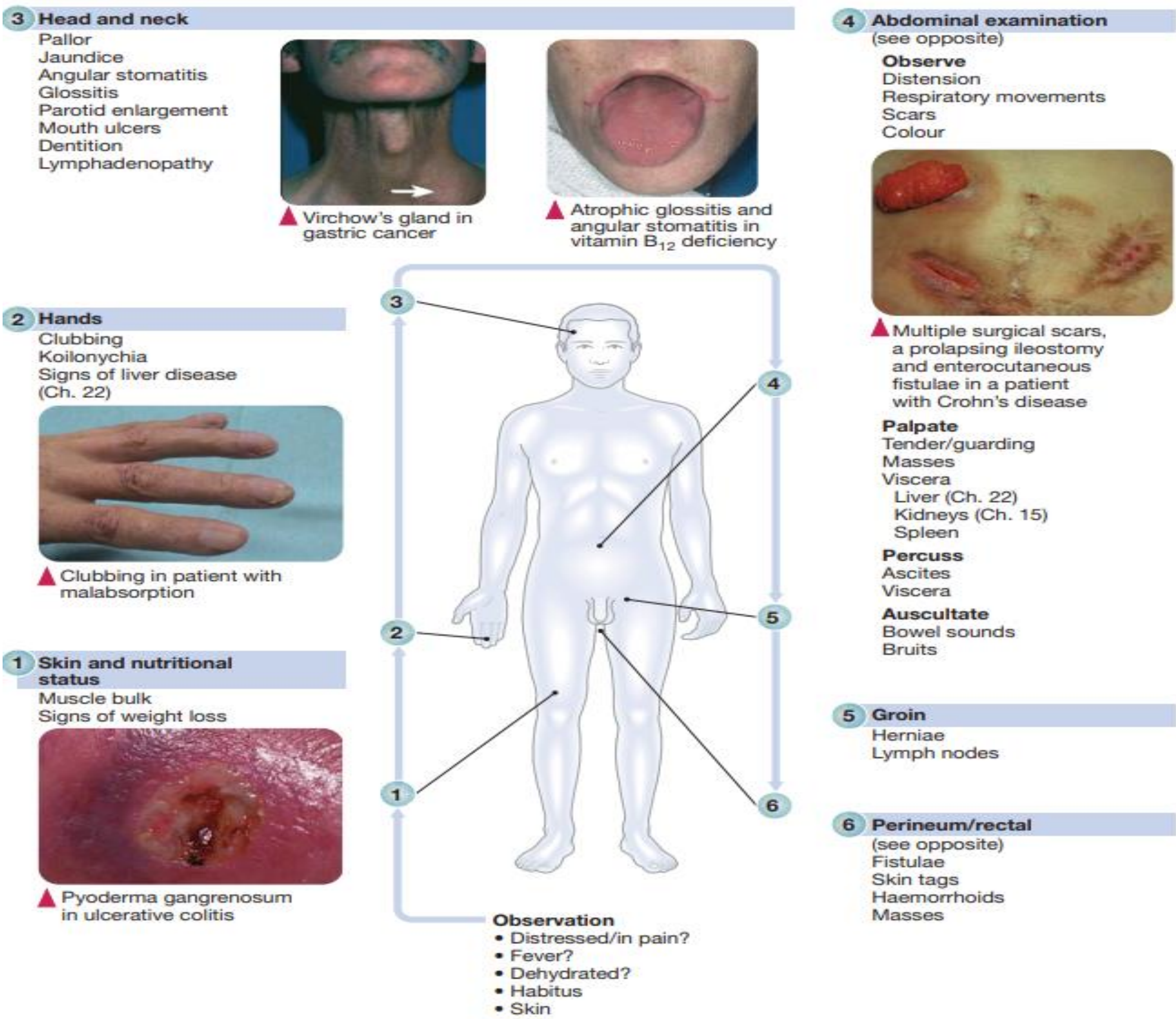
By the end of this lecture the student will be able to :

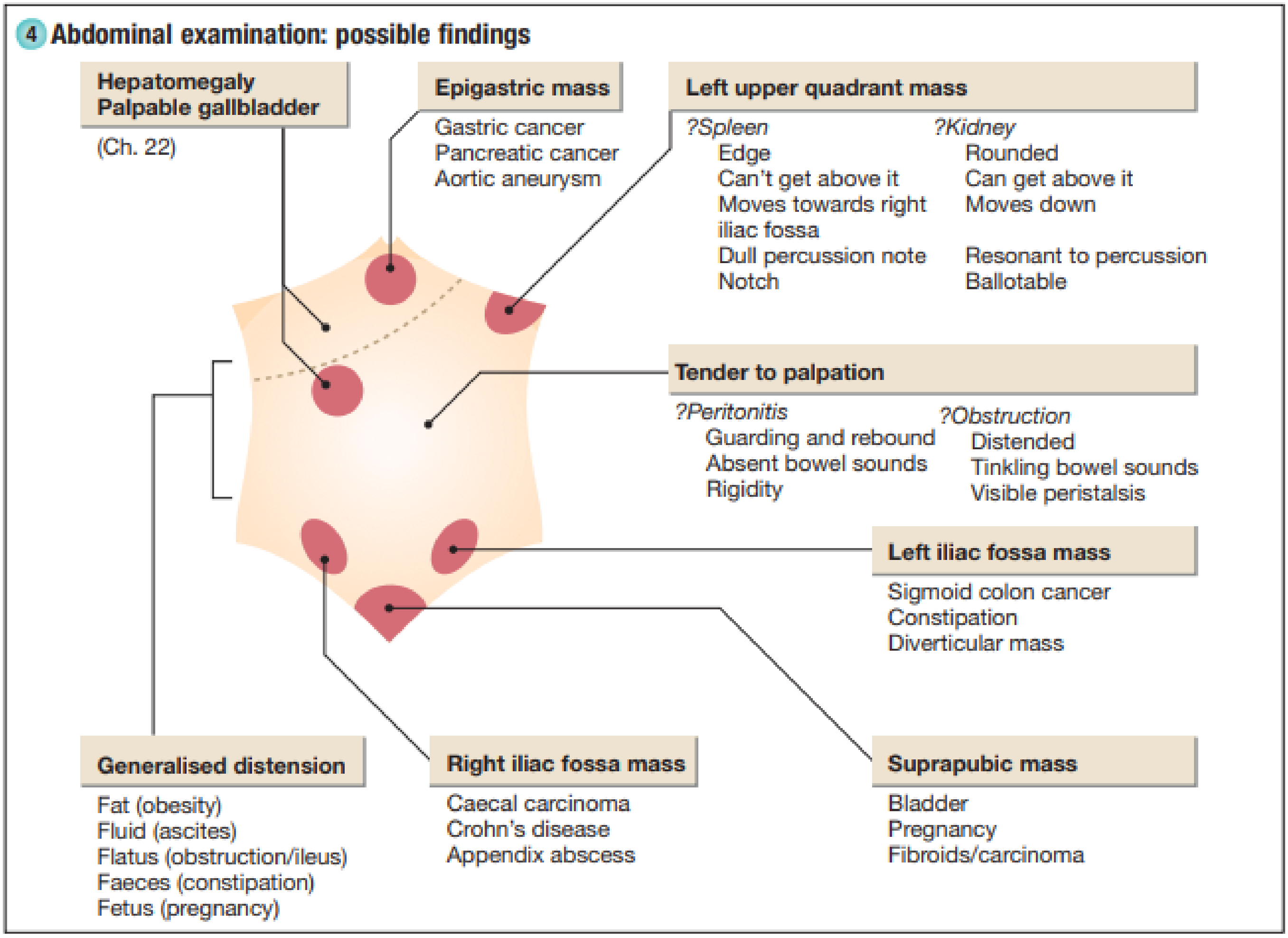
1 .Identify the anatomy of the gastrointestinal tract

2 .Explain the physiology of the digestive tract .

3 .To know the hormones & enzymes of the GIT system .

Clinical examination of the gastrointestinal tract





Significance of the GI system

- Diseases of the gastrointestinal tract are a major cause of morbidity and mortality.
- Around 10% of all consultations in primary care are for gastroenterological problems, split equally between the upper and lower gastrointestinal tract.
- **Infective diarrhoea** and **malabsorption** are responsible for much ill health and many deaths in low- and middle-income countries.
  The gastrointestinal tract is the most common site for cancer development.
  Colorectal cancer is the third most common cancer in men and second most common in women.
- Functional gastrointestinal disorders account for at least **40%** of all referrals to gastroenterology services and consume considerable health-care resources .

Functional anatomy & physiology

Oesophagus & stomach :

The **Oesophagus** is a **muscular tube** that extends **25 cm** from the **cricoid cartilage** to the **cardiac orifice** of the stomach, with the **gastroesophageal junction** being **40 cm from the incisors**. The oesophagus has both an **upper** and a **lower sphincter**. A **peristaltic swallowing wave** propels the **food bolus** into the **stomach**.
The **stomach** acts as a '**hopper**', **retaining** and **grinding food**, and then actively propelling it into the **upper small bowel**

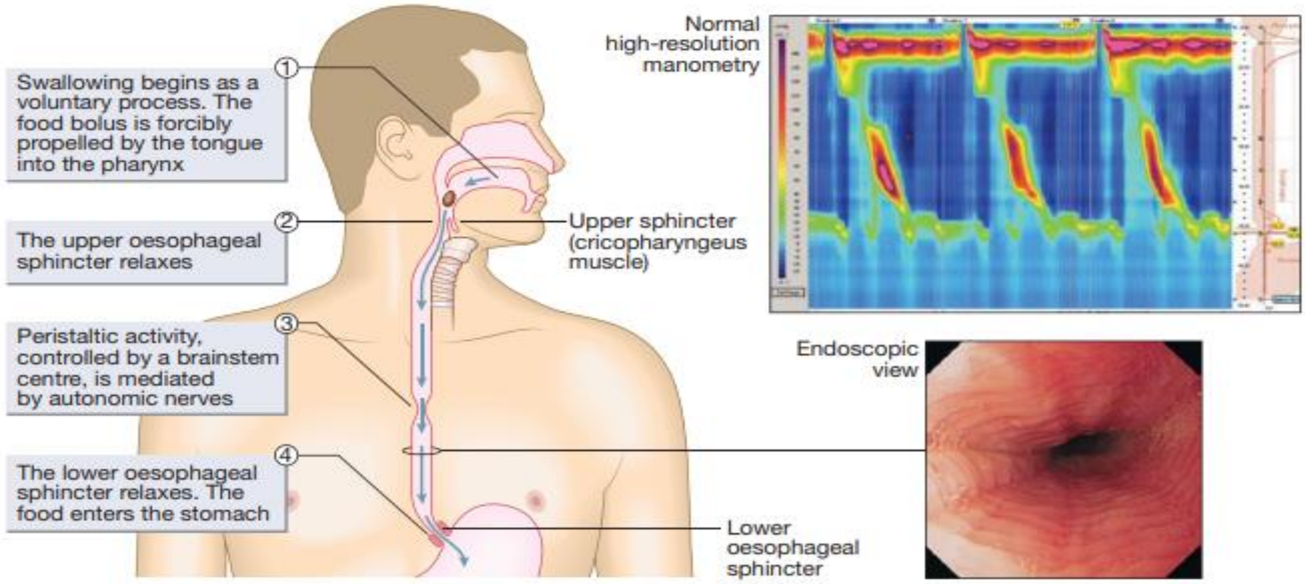


Fig. 21.1 The oesophagus: anatomy and function. The swallowing wave.

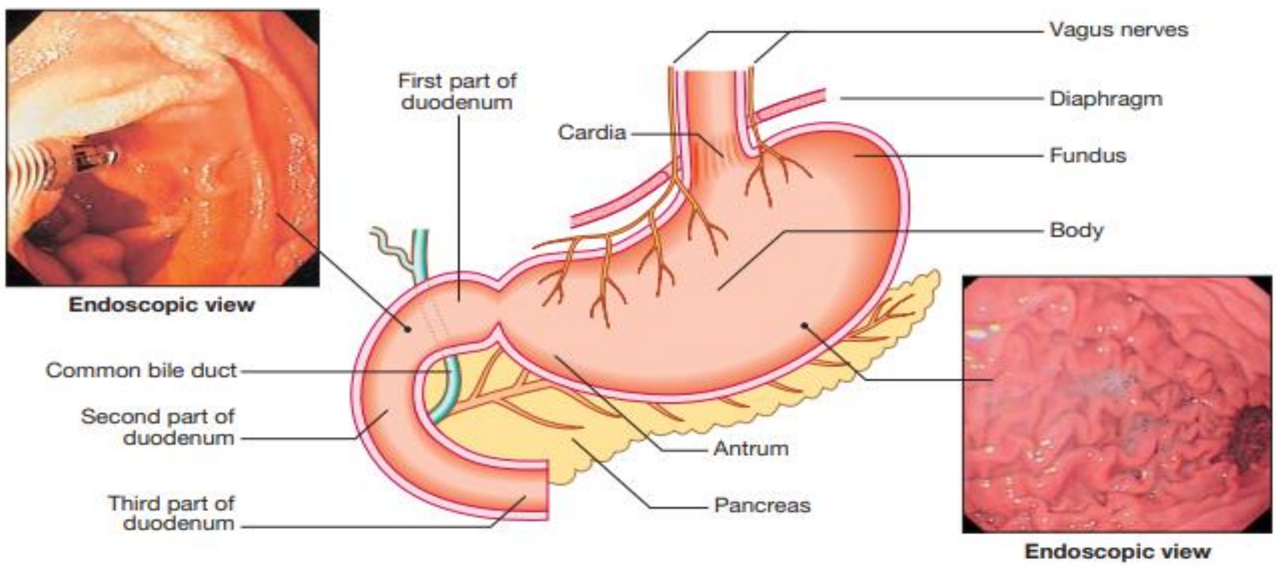


Fig. 21.2 Normal gastric and duodenal anatomy.

Gastric secretion

The **hormone gastrin** is produced by **G cells** mainly in the **gastric antrum**, with **somatostatin** being secreted from **D cells** throughout the **gastrointestinal tract**. **Gastrin** stimulates **acid secretion** and **mucosal growth** while **somatostatin** suppresses it. **Ghrelin**, secreted from **oxyntic glands**, stimulates **acid secretion**, as well as **appetite** and **gastric emptying**.

Bicarbonate ions, stimulated by **prostaglandins**, **mucins**, and **trefoil factor family (TFF) peptides**, together protect the **gastroduodenal mucosa** from the **ulcerative properties** of **acid** and **pepsin**.

Small intestine:

The **jejunum** extends from the **ligament of Treitz** to the **ileocaecal valve**. During **fasting**, a wave of **peristaltic small bowel** occurs every **1–2 hours**. Entry of **food** into the **gastrointestinal tract** stimulates **small bowel peristaltic activity**.

Functions of the small intestine are :

- **Digestion** (mechanical, enzymatic, and peristaltic)
- **Absorption** – the products of digestion, **water**, **electrolytes**, and **vitamins**
- **Protection** against **ingested toxins.**
- **Immune regulation.**

Protective function of the small intestine:

A. **Physical defense mechanisms**
B. **Immunological defense mechanisms**

A. Physical defense mechanisms:

There are several levels of defense in the small bowel. Firstly, the **gut lumen** contains **host bacteria**, **mucins**, and secreted **antibacterial products**, including **defensins** and **immunoglobulins**, that help combat **pathogenic infections**. Secondly, **epithelial cells** have relatively **impermeable brush border membranes**, and passage between cells is prevented by **tight** and **adherens junctions**.
Lastly, in the **subepithelial layer**, **immune responses** occur under the control of the **adaptive immune system** in response to **pathogenic compounds**.

B. **Immunological defense mechanisms**:

- **Gastrointestinal mucosa-associated lymphoid tissue (MALT)** constitutes **25%** of the total **lymphatic tissue** of the body and is at the heart of **adaptive immunity**.
- Within **Peyer's patches**, **B lymphocytes** differentiate into **plasma cells** following exposure to **antigens**, and these migrate to **mesenteric lymph nodes** to enter the bloodstream via the **thoracic duct**.
- The **plasma cells** return to the **lamina propria** of the gut through the circulation and release **immunoglobulin A (IgA)**, which is transported into the **lumen** of the intestine.
- **Intestinal T lymphocytes** help localize **plasma cells** to the site of **antigen exposure**, as well as producing **inflammatory mediators**. **Macrophages** in the gut release a range of **cytokines**, which mediate **inflammation**.

Colon:

The **colon** absorbs **water** and **electrolytes**. It also acts as a **storage organ** and has **contractile activity**. Two types of contraction occur. The first is **segmentation** (**ring contraction**), which leads to **mixing** but not propulsion; this promotes the absorption of **water** and **electrolytes**. **Propulsive** (**peristaltic contraction**) waves occur several times a day and propel **faeces** to the **rectum**. All activity is stimulated after meals through the **gastrocolic reflex** in response to the release of hormones such as **5-hydroxytryptamine (5-HT, serotonin)**, **motilin**, and **CCK**.

Intestinal microbiota

There are many **environmental factors** that can affect the **microbiota**, including **diet**, **drugs**, **physical activity**, **smoking**, **stress**, and **natural ageing**. Generally, the **adult intestinal microbiota** is acquired by the age of **2 years**. A **dysbiosis** or imbalance between the different components of the intestinal microbiota has been associated with diseases of the gastrointestinal tract, such as **inflammatory bowel disease** and **colorectal cancer**; **liver disease**, including **hepatocellular carcinoma**; and pathologies outside the gastrointestinal tract, such as **diabetes**, **obesity**, **cardiovascular disorders**, **cerebrovascular disorders**, **asthma**, and **psychiatric disorders**, such as **depression**. Many challenges remain in understanding the **intestinal microbiota** and how it impacts **health and disease**.

The nervous system and gastrointestinal function:

The **central nervous system (CNS)**, the **autonomic nervous system (ANS)**, and the **enteric nervous system (ENS)** interact to regulate **gut function**.
The ANS comprises:

- **Parasympathetic pathways** (**vagal** and **sacral efferent**), which are **cholinergic**, increase **smooth muscle tone**, and promote **sphincter relaxation**.
- **Sympathetic pathways**, which release **noradrenaline (norepinephrine)**, reduce **smooth muscle tone**, and stimulate **sphincter contraction**.

21.2 Gut hormones and peptides

Hormone	Origin	Stimulus	Action
Gastrin	Stomach (G cell)	Products of protein digestion Suppressed by acid and somatostatin	Stimulates gastric acid secretion Stimulates growth of gastrointestinal mucosa
Somatostatin	Throughout gastrointestinal tract (D cell)	Fat ingestion	Inhibits gastrin and insulin secretion Decreases acid secretion Decreases absorption Inhibits pancreatic secretion
Cholecystokinin (CCK)	Duodenum and jejunum (I cells); also ileal and colonic nerve endings	Products of protein digestion Fat and fatty acids Suppressed by trypsin	Stimulates pancreatic enzyme secretion Stimulates gallbladder contraction Relaxes sphincter of Oddi Modulates satiety Decreases gastric acid secretion Reduces gastric emptying Regulates pancreatic growth
Secretin	Duodenum and jejunum (S cells)	Duodenal acid Fatty acids	Stimulates pancreatic fluid and bicarbonate secretion Decreases acid secretion Reduces gastric emptying
Motilin	Duodenum, small intestine and colon (Mo cells)	Fasting Dietary fat	Regulates peristaltic activity, including migrating motor complexes (MMCs)
Gastric inhibitory polypeptide (GIP)	Duodenum (K cells) and jejunum	Glucose and fat	Stimulates insulin release (also known as glucose-dependent insulinotrophic polypeptide) Inhibits acid secretion Enhances satiety
Glucagon-like peptide-1 (GLP-1)	Ileum and colon (L cells)	Carbohydrates, protein and fat	Stimulates insulin release Inhibits acid secretion and gastric emptying Enhances satiety
Vasoactive intestinal peptide (VIP)	Nerve fibres throughout gastrointestinal tract	Unknown	Has vasodilator action Relaxes smooth muscle Stimulates water and electrolyte secretion
Ghrelin	Stomach	Fasting Inhibited by eating	Stimulates appetite, acid secretion and gastric emptying
Peptide YY	Ileum and colon	Feeding	Modulates satiety

Reference: Davidson''s principle & practice of medicine.

Summery

The gastrointestinal system plays a crucial role in health, impacting overall well-being and disease. GI diseases lead to significant health problems and deaths, with around 10% of doctor visits related to stomach issues. In low-income countries, diarrhea and malabsorption are major health concerns, while colorectal cancer is among the most common cancers.

The GI tract includes the oesophagus and stomach, where the oesophagus moves food to the stomach, which grinds and holds it. Key hormones like gastrin promote acid production, while somatostatin reduces it. Ghrelin stimulates appetite, and bicarbonate helps protect the stomach lining.

The small intestine is vital for digestion, absorption of water and nutrients, and immune protection. It has both physical defenses (like mucus and bacteria) and immunological defenses through special tissues that help fight infections.

The colon absorbs water and stores waste, using contractions to mix and move feces, especially after meals. The balance of gut bacteria is also important; factors like diet and stress can disrupt this balance, leading to diseases such as diabetes and mental health issues.

Finally, the nervous system—comprising the central nervous system (CNS), autonomic nervous system (ANS), and enteric nervous system (ENS)—works together to regulate gut function. The ANS includes pathways that help control muscle relaxation and contraction in the gut.